

1. Combine the Effects

A meta-analysis was conducted to combine the results of 27 studies examining differences between boys and girls in play with stereotypically male toys. Boys were treated as the experimental group and girls as the control group.

The random effects model showed a statistically significant positive effect size ($MD = 94.78$, $p < 0.0001$). This indicates that boys spent significantly more time playing with stereotypically male toys compared to girls.

The heterogeneity between studies was very high ($I^2 = 88\%$), suggesting that the included studies differed substantially in their reported effect sizes. Therefore, the random effects model was considered more appropriate than the common/fixed effect model.

The forest plot showed that most studies reported positive effects, although the magnitude of the effects varied considerably across studies.

2. Funnel Plot and Publication Bias

A funnel plot was created to examine potential publication bias and small-study effects.

The funnel plot appeared asymmetric, with more studies located on the positive side of the pooled effect size. Smaller studies also tended to report larger positive effects.

This asymmetry may suggest possible publication bias or small-study effects. However, the high heterogeneity observed in the meta-analysis may also contribute to the asymmetry of the funnel plot. Therefore, the funnel plot should be interpreted with caution.

Overall, most studies indicated that boys spent more time playing with stereotypically male toys than girls.

3. Do Methods or Quality Affect the Results?

Meta-regression analysis was used to examine whether methodological variables affected the reported effect sizes. The following moderators were tested: neutral toys, parent presence, setting, country, and year.

The overall moderator test was statistically significant ($QM\ p = 0.0035$), suggesting that at least one methodological variable affected the reported effect sizes.

Among the tested moderators, Year was statistically significant and had a negative coefficient. This indicates that more recent studies tended to report smaller differences between boys and girls in play with stereotypically male toys.

Neutral toys, parent presence, setting, and country were not statistically significant moderators.

The moderators explained approximately 40.5% of the heterogeneity between studies.

Quality Effects

A separate meta-regression tested whether study quality, measured using the NOS score, affected the reported effect sizes.

The NOS score showed a marginally significant negative relationship with effect sizes ($p = 0.0548$). Higher-quality studies tended to report smaller differences between boys and girls in play with stereotypically male toys.

Study quality explained approximately 12.8% of the heterogeneity between studies.

4. Does Author Gender Affect the Results?

A meta-regression analysis was conducted using the proportion of female authors as a moderator.

The results showed that author gender composition was not a statistically significant moderator ($p = 0.7416$). This suggests that the proportion of female and male authors did not meaningfully affect the reported differences between boys and girls in play with stereotypically male toys.

Author gender explained virtually none of the heterogeneity between studies.

5. Conclusion

The meta-analysis found strong evidence that boys spend more time playing with stereotypically male toys than girls. However, there was substantial heterogeneity between studies, indicating that study results varied considerably.

The funnel plot suggested possible publication bias or small-study effects, although the high heterogeneity makes interpretation difficult.

Methodological factors, especially study year, influenced the reported effect sizes, with more recent studies reporting smaller gender differences. Study quality also appeared related to effect sizes, although the relationship was only marginally significant.

Finally, author gender composition did not significantly affect the reported results.